

Saad Shahd

Principal Software Engineer

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Principal Software Engineer building real-time distributed systems that scale. I make architectural decisions that unlock velocity while maintaining correctness—through event-driven patterns, immutable data flows, and infrastructure designed to fail gracefully. 12+ years taking platforms from prototype to production at companies like Wise and StatsBomb, handling millions of events per second.

Key Expertise

Distributed Systems & High-Scalability Infrastructure Design

Architecting systems that handle millions of events while maintaining correctness guarantees and operational simplicity.

Real-Time Data Stream Processing (Kafka)

Building event-driven architectures with immutable streams, exactly-once semantics, and graceful degradation.

Microservice Architecture (DDD, CQRS, gRPC)

Decomposing monoliths into bounded contexts with clear interfaces, reducing coupling while maintaining consistency.

Platform Engineering & Infrastructure Evolution

Designing infrastructure that lets teams deploy confidently—through observability, automated testing, and progressive delivery.

Experience

Senior Software Engineer

London | 03/2022 - Present

Wise

Building infrastructure for the world's most international account—helping 16M+ customers move money across borders.

Increased conversion 19% through hypothesis-driven experimentation. Isolated pricing friction points through rapid A/B testing. Short iterations revealed psychological barriers traditional analytics missed—validating hunches with data, not opinions.

Improved team velocity 4x by eliminating accidental complexity. Refactored tangled service dependencies into composable modules with clear boundaries. Each team could now deploy independently. Reduced coordination overhead from "everyone talks to everyone" to "services talk through contracts."

Increased user satisfaction 26% by consolidating fragmented product components. Migrated six inconsistent UIs into a unified design system backed by shared state management. Users no longer learned different interactions for similar actions—cognitive load dropped, satisfaction rose.

Principal Software Engineer

Bath, UK | 03/2018 - 03/2022

StatsBomb

Led core data infrastructure for football analytics platform—serving Premier League clubs, international federations, and media companies.

Architected real-time data streaming platform from scratch, handling 10K+ events/second. Chose event sourcing with Kafka for immutable audit trails—critical when clubs make million-pound decisions on our data. Event-driven design let us replay history, debug production issues in development, and maintain consistency across services without distributed transactions.

Increased data collection speed 6x through parallel processing architecture. Previous sequential pipeline became the bottleneck—one slow collector blocked everyone. Redesign as independent stream processors with coordinated checkpoints. Tradeoff: More complex failure handling, but throughput scaled linearly with workers.

Improved data quality 8x by making validation impossible to skip. Moved from "trust but verify" post-processing to typed schemas with compile-time guarantees. Invalid data couldn't enter the system—structural correctness over runtime checks. Quality issues dropped from daily firefighting to rare edge cases.

These platform improvements drove 3x increase in client satisfaction. Faster turnaround meant clubs received insights before match deadlines. Better quality meant fewer "wait, this can't be right" conversations. Technical decisions directly impacted renewal rates—architecture has business consequences.

Senior Software Engineer

Cairo, Egypt | 03/2017 - 03/2018

Eventtus

Built engagement platform for events—the leading event app provider across Middle East and North Africa.

Architected entire platform as decoupled microservices using DDD, CQRS, and gRPC. Decomposed monolithic Rails app into bounded contexts—attendees, sessions, networking, analytics. Each context owned its data, exposed typed interfaces, communicated through events. Teams could evolve services independently without coordination tax. Pattern: Eventual consistency over distributed transactions—accept temporary staleness for operational simplicity.

Senior Frontend Engineer

Cairo, Egypt | 11/2015 - 08/2016

Instabug

Core team member building tools for mobile developers to ship better apps—crash reporting, user feedback, and performance monitoring.

Built real-time crash reporting dashboard processing 100K+ events daily. Designed client SDK for minimal performance overhead while capturing rich debugging context. Challenge: Collect enough data to reproduce bugs without impacting app performance—solved through smart sampling, local aggregation, and batched uploads.

Skills

Architecture & Systems

Expert: Distributed Systems, Event-Driven Architecture, Microservices, DDD, CQRS

Proficient: System Design, Platform Engineering, Real-Time Processing, Stream Processing

Backend & Data

Expert: Node.js, Kafka, Redis, SQL, REST, GraphQL, gRPC

Proficient: Python, Rust, Clojure, Scala, RocksDB, MongoDB, Neo4j, Druid

DevOps & Infrastructure

Expert: Kubernetes, Docker, AWS, Prometheus, Grafana

Proficient: Helm, JenkinsX, GCP, Nginx, HAProxy

Frontend

Expert: TypeScript, React, HTML/CSS

Proficient: Vue, Angular, Redux, xstate, RxJS, D3, Electron

Find Me Online

GitHub: github.com/saadshahd

LinkedIn: linkedin.com/in/saadshahd